

PROJECT PROPOSAL

IT138IU – Data Science and Data Visualization

Resource and Energy Sustainability: Vietnam vs. 7 Southeast Asian Countries

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Repository: <https://github.com/LeXuanQue/sea-resource-sustainability-viz>

1. Background and motivation

The World Bank's Sovereign ESG framework evaluates the environmental dimension of economic performance through natural-resource endowment and management alongside sustainable energy use [1], which makes forest resources, resource depletion, freshwater use, energy intensity, renewable energy and fossil-fuel dependence relevant dimensions of environmental sustainability. For Viet Nam this is also national policy: the National Climate Change Strategy targets net-zero greenhouse-gas emissions by 2050 and a 2050 forest cover target of 43% [2]. The World Bank records Viet Nam's forest area at 47.2% of land area in 2022 [1]; the two figures use different forest definitions and are not directly comparable.

According to Ember, Viet Nam's electricity demand grew by an average of 8.2% per year between 2013 and 2024, and coal-based generation expanded over the same period to meet it, while wind and solar also grew substantially [3]; across the region, coal supplied 45% of ASEAN electricity generation in 2024 [3]. Viet Nam alone therefore cannot show its position, whereas comparing it with Indonesia, Thailand, Malaysia, the Philippines, Cambodia, Laos and Myanmar reveals differences in resource conditions, resource pressure and energy transition without assuming one indicator represents sustainability as a whole.

A static report suits this poorly: an interactive interface lets users pick an indicator, year or country and read the trend and regional comparison directly, rather than producing a single composite sustainability score.

2. Objectives

This project aims to develop an interactive web-based visualization for exploring natural-resource and energy sustainability indicators across Viet Nam and seven selected Southeast Asian countries. The visualization will enable users to examine the historical development of 12 indicators, compare Viet Nam with the selected regional countries, and observe how Viet Nam's relative position changes over time. The project focuses on supporting data exploration and comparison rather than producing a single composite sustainability score. Specifically, the interactive visualization is designed to answer the following research questions:

- **RQ1:** How have Viet Nam's 12 natural-resource and energy indicators changed over time?
- **RQ2:** How does Viet Nam compare with Indonesia, Thailand, Malaysia, the Philippines, Cambodia, Laos and Myanmar across the 12 indicators over time?
- **RQ3:** How does the gap between Viet Nam and the selected-country regional median change over time for the 12 indicators?

3. Data

Data source: One public source, the World Bank Sovereign ESG dataset [1], downloaded once as a CSV (`WB_ESG_WIDEF.csv`, 239 economies, 71 core indicators, 1960–2023), so no scraping or API is needed and the dataset is fixed and reproducible.

Scope: 8 countries (Vietnam, Indonesia, Thailand, Malaysia, the Philippines, Cambodia, Laos and Myanmar) and 12 indicators from two Environment themes, all present for all 8 countries, with coverage checked row by row. Indicator codes are listed in the repository README.

Group	Indicator	Unit	Years
Resources	Forest area	% of land area	1990–2022
Resources	Tree cover loss	hectares	2002–2021
Resources	Protected areas	% of territory	2013–2023
Resources	Resource depletion	% of GNI	1990–2021
Resources	Net forest depletion	% of GNI	1990–2021
Resources	Freshwater withdrawals	% of internal res.	1990–2021
Energy	Energy intensity	MJ/\$2017 PPP GDP	2000–2022
Energy	Renewable energy	% of final energy	1990–2022
Energy	Renewable electricity	% of electricity	1990–2021
Energy	Energy use per person	kg oil equivalent	1990–2022
Energy	Fossil fuel use	% of total energy	1990–2022
Energy	Electricity from coal	% of electricity	1990–2022

Coverage: Uneven for four countries: Cambodia starts in 1995 on most series and Laos in 2000 on energy use, fossil fuel and coal; freshwater withdrawals start in 2005–2007 for Laos, the Philippines, Cambodia and Thailand; and Laos reports renewable electricity for only 10 scattered years between 2001 and 2021.

4. Data processing

A Python script turns the wide CSV into three static JSON files the site reads directly; nothing is computed in the browser except filtering.

- **Filter:** the 8 countries, the 12 indicators above, years 1990–2023.
- **Reshape:** wide to long form, one row per country, indicator, year and value.
- **Keep gaps as gaps:** missing values are stored as null, never 0, and charts never interpolate across them.
- **Flag repeated values:** Vietnam's freshwater withdrawals sit at exactly 22.78% from 2005 to 2021; such runs are drawn lighter.
- **Choose the year shown:** the latest year with at least 6 of the 8 countries reporting, printed beside every value.
- **Regional reference:** the median of the countries reporting that year, since one country can drag the mean far (freshwater: mean 11.0%, median 7.4%).
- **Direction and rank:** indicators are tagged higher- or lower-is-better and ranks direction-adjusted, so rank 1 is always best; energy use per person has no agreed direction and no rank.
- **Verdict:** better or worse by position against the median, level within 3% of it; change since 2010 is judged separately.
- **Export:** values.json (long form), snapshot.json (latest value, median, rank) and coverage.json (first year, last year, gaps).

Known limitations: tree cover loss is in hectares, so larger countries look worse by construction; protected areas has no 2010 baseline; values are national averages.

5. Visualization design

All three designs answer the same question with the same three levels (region, one country, one indicator over time) and differ in how the reader moves between them. The six sketches are in the appendix.

5.1 Design A: story first, then explore

The page opens as a short explanation and only then hands control to the reader, following the "martini glass" structure of narrative visualization [4] (Figure 1). A hero states the question, a one-sentence answer and three headline numbers with a better or worse tag: forest area +18.4 points since 1990, renewable energy share -51.7 points since 1990, and 2.9% of territory protected. Three story blocks follow, each with a conclusion as its title, one chart and two sentences. An explore section then gives shared indicator, country and year controls driving a ranked bar list with an orientation map, a dot plot of all 12 indicators for the selected country, and a trend line against the regional median; clicking a country drills down and a back control returns (Figure 2). Only Vietnam is coloured, and better and worse carry symbol, word and colour together.

Strengths: a reader who never touches a control still gets the main message, and every chart uses position or length. **Weaknesses:** the page is long, and the story blocks must be rewritten if the data changes.

5.2 Design B: coordinated dashboard

Everything sits on one screen (Figure 3): a top bar with shared indicator, country and year controls, a KPI strip of six cells under "Natural resources" and "Energy" tabs, a ranked bar list of the 8 countries with the regional median dashed across it, six small multiples, and one large trend chart. Clicking a country switches every panel together, with a removable chip showing the selection (Figure 4). **Strengths:** every level is visible at once. **Weaknesses:** the main message needs interaction, and six panels is dense.

5.3 Design C: head-to-head comparison

Three full-page levels sit inside one shell with a fixed 01, 02, 03 header. Level 1 pairs a regional map with a ranked bar list and cards for the regional average and median (Figure 5); level 2 shows all 12 indicators as sparkline cards for one country against a comparison country; level 3 is a multi-line time-series chart with country chips, a year cursor and a coverage table. Clicking a country opens level 2 and an indicator card opens level 3 (Figure 6). **Strengths:** the most explicit path between levels, with missing data labelled throughout, including named gaps for Cambodia before 2005 and Laos before 2002. **Weaknesses:** it benchmarks against the regional average, conflicting with our median rule, and it is the largest build.

5.4 Chosen direction

Criterion	Design A	Design B	Design C
Main message readable without interaction	Yes	No	Partly
Three levels with a path between them	Drill-down and back	All on one screen	Most explicit
Shows missing data explicitly	Yes	Yes	Yes
Build effort in 10 weeks, no prior frontend	Medium	Medium to high	High

Chosen direction: we will implement Design A. It is the only one of the three whose main finding is readable with no interaction, and it still contains the full explore section, so it answers every research question while staying the lowest-risk build for a team with no prior frontend experience.

6. Must-have features

What the week-6 prototype needs to answer the research questions.

- **M1 Region level:** ranked bar list of the 8 countries for one indicator and year, regional median marked.
- **M2 Country level:** all 12 indicators for one country, each with its rank among the 8 and its verdict against the median.
- **M3 Indicator level:** trend line from 1990, selected country highlighted, regional median dashed.

- **M4 Shared controls:** indicator selector, country selector and year slider, every panel updating together.
- **M5 Drill-down:** clicking a country in the ranking opens its country view; a back control returns.
- **M6 Tooltips:** country, year, value and unit on every mark.
- **M7 Missing data:** shown explicitly as "no data", line breaks or hatching, never as zero.
- **M8 Provenance:** units on every chart and the World Bank Sovereign ESG source on the page.
- **M9 Delivery:** static site on GitHub Pages, no backend.
- **M10 Story section:** hero with three headline findings and three story blocks, readable without interaction.

Every research question is answered by at least one must-have feature: RQ1 by M3, M2 and M10, RQ2 by M1, M4 and M5, and RQ3 by M3, M2 and M10.

7. Optional features

Added only if time allows after week 7.

- Orientation map of Southeast Asia beside the ranking.
- Collapsible details section with the indicator table and per-country coverage.
- Animated transitions when the year or indicator changes.
- Shareable links that keep the current selection in the URL.
- Vietnamese and English versions of the interface.

8. Project schedule

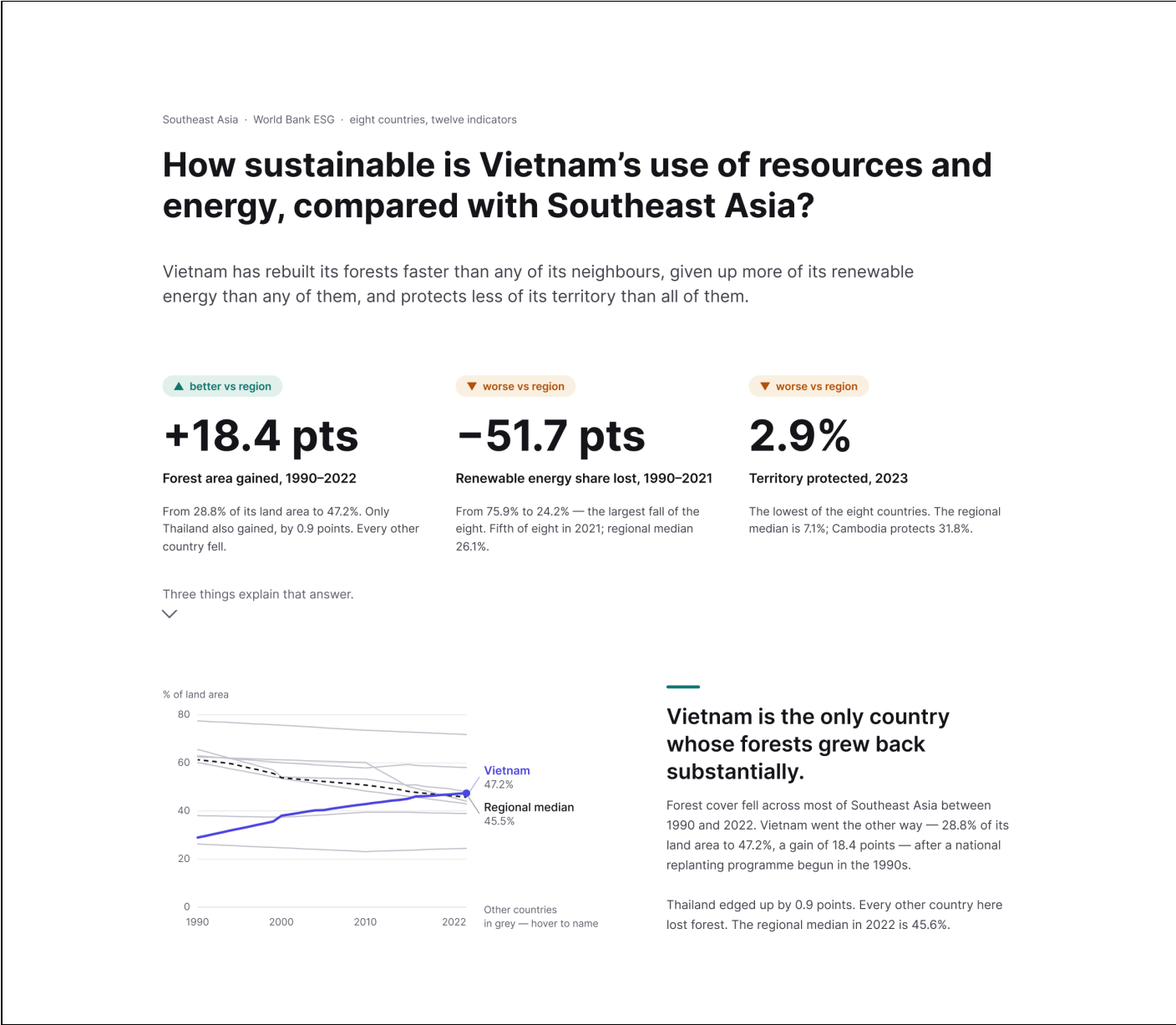
Ten weeks from Monday 5 October 2026; the week-by-week plan is in the repository at `docs/project-schedule.md`.

Week	Milestone
1–2	Kickoff, proposal, design choice, JSON schema, page skeleton
3–4	JSON v1; region, country and indicator levels; shared controls; drill-down
5–6	Midterm integration and bug fixes; working prototype due (week 6)
7–8	Feedback, improvements, optional features, report sections
9–10	Feature freeze, testing, report; final submission (week 10)

Weeks 3–4 are led by Trần Nguyễn Lê Quân (data), Lê Xuân Quế and Đặng Hoàng Quân (frontend); all other weeks are shared. Weeks 1–2 include time to learn D3.js, as no member has built a frontend before.

References

- [1] World Bank. Sovereign ESG Data Framework. <https://esgdata.worldbank.org/data/framework>, accessed 2 October 2026.
- [2] Chính phủ Việt Nam. 2022. Chiến lược quốc gia về biến đổi khí hậu đến năm 2050 (National Climate Change Strategy to 2050). <https://xaydungchinhhsach.chinhphu.vn/phe-duyet-chien-luoc-quoc-gia-ve-bien-doi-khi-hau-den-nam-2050-119220727070101019.htm>, accessed 2 October 2026.
- [3] Ember. From emission-intensive to investment hotspots: championing renewables in 3 ASEAN economies. <https://ember-energy.org/latest-insights/from-emission-intensive-to-investment-hotspots-championing-renewables-in-3-asean-economies/country-snapshots-and-opportunities/>, accessed 2 October 2026.
- [4] Segel, E. and Heer, J. 2010. Narrative Visualization: Telling Stories with Data. *IEEE Transactions on Visualization and Computer Graphics* 16(6).



Click a country (drill-down)

Clicking Laos switches every panel to Laos. Vietnam stays on screen as a reference. Renewable electricity also shows the gap rule: Laos reports ten scattered years, and the isolated 2012 reading is a single dot.

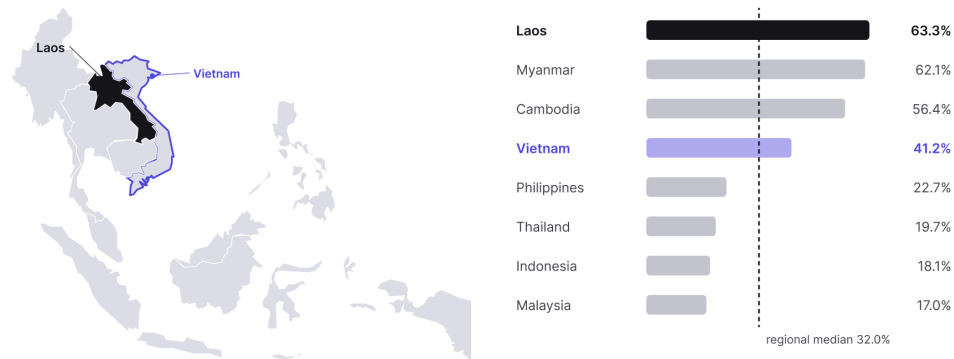
Now explore the rest yourself.

One indicator, one country, one year — shared by all three panels. The year starts at the latest the indicator reports.

Indicator **Renewable electricity** Country **Laos** Year **2021** 1990 2021 [← Back to Vietnam](#)

Laos gets the largest share of its electricity from renewables.

% of electricity, 2021 · best at top · dashed line is the median of the 8 countries reporting



The map is here for orientation — the values are in the list.

↓ click a country

Drag the year to 1995

Every panel follows the year. In 1995 four countries have no freshwater reading, and three whole indicators do not start until later.

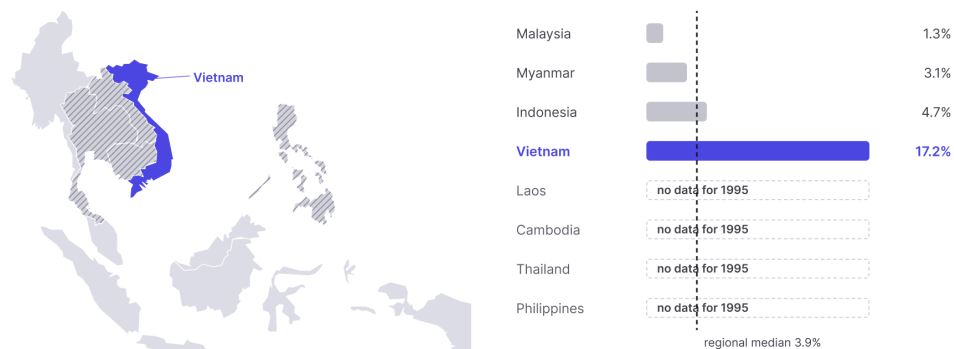
Now explore the rest yourself.

One indicator, one country, one year — shared by all three panels. The year starts at the latest the indicator reports.

Indicator **Freshwater withdrawals** Country **Vietnam** Year **1995** 1990 2021

Only four of the eight countries reported freshwater withdrawals in 1995.

% of internal resources, 1995 · best at top · dashed line is the median of the 4 countries reporting



The map is here for orientation — the values are in the list.

Figure 2. Design A: clicking a country re-ranks and highlights it; years with no data are labelled, never drawn as zero.

SEA Sustainability

Vietnam vs 7 Southeast Asian countries · World Bank Sovereign ESG

INDICATOR

Forest area (% of land area)

COUNTRY

Vietnam

YEAR

1990 2022

Natural resources

Energy

vs regional median · latest year per indicator

Forest area

47.2

% of land area

▲ better

median 45.6 · 2022

Tree cover loss

193k

hectares

▲ better

median 235k · 2021

Protected areas

2.9

% of territory

▼ worse

median 7.05 · 2023

Resource depletion

0.51

% of GNI

▲ better

median 2.31 · 2021

Net forest depl.

0

% of GNI

= level

median 0 · 2021

Freshwater use

22.8

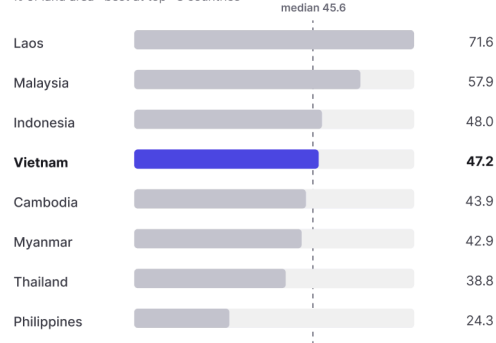
% of internal resources

▼ worse

median 7.44 · 2021

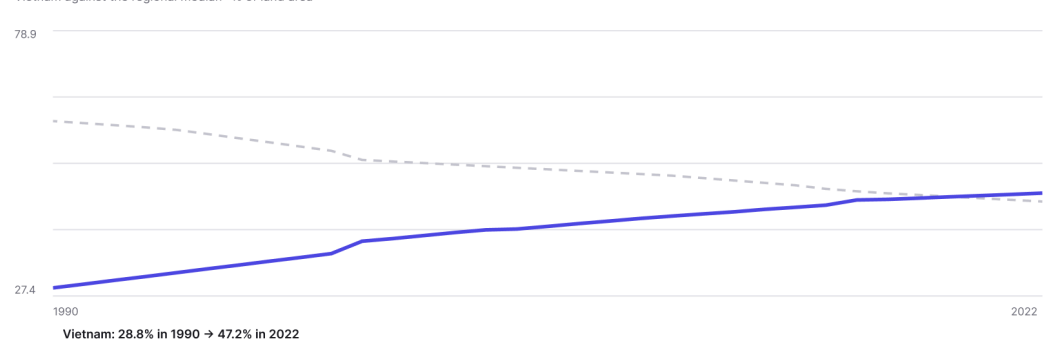
Forest area, 2022

% of land area · best at top · 8 countries



Forest area over time, 1990–2022

Vietnam against the regional median · % of land area

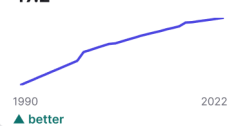


ALL SIX NATURAL-RESOURCE INDICATORS FOR VIETNAM

Forest area

47.2

% of land area



Tree cover loss

193k

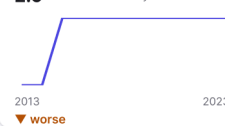
hectares



Protected areas

2.9

% of territory



Resource depletion

0.51

% of GNI



Net forest depl.

0

% of GNI



Freshwater use

22.8

% of internal resources



Regional reference is the MEDIAN of the 8 countries, never the mean. Missing values are shown as "no data", never 0. Ranks are direction-adjusted, so rank 1 is always the best outcome.

Source: World Bank Sovereign ESG · esgdata.worldbank.org

Figure 3. Design B: coordinated dashboard, every view on one screen.

SEA Sustainability

Vietnam vs 7 Southeast Asian countries · World Bank Sovereign ESG

INDICATOR

Forest area (% of land area)

COUNTRY

Laos

YEAR

1990 2022

Natural resources

Energy

vs regional median · latest year per indicator

Laos selected x

Forest area

71.6

% of land area

▲ better

median 45.6 · 2022

Tree cover loss

318k

hectares

▼ worse

median 235k · 2021

Protected areas

19.0

% of territory

▲ better

median 7.05 · 2023

Resource depletion

3.44

% of GNI

▼ worse

median 2.31 · 2021

Net forest depl.

1.59

% of GNI

▼ worse

median 0 · 2021

Freshwater use

3.86

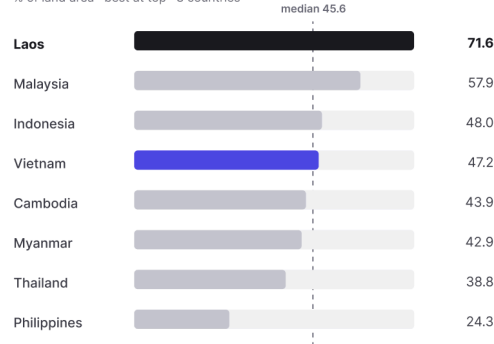
% of internal resources

▲ better

median 7.44 · 2021

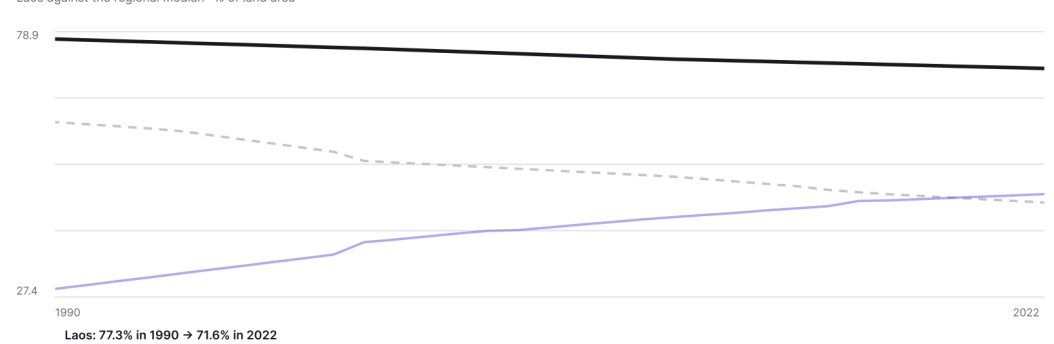
Forest area, 2022

% of land area · best at top · 8 countries



Forest area over time, 1990-2022

Laos against the regional median · % of land area

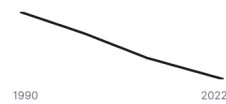


ALL SIX NATURAL-RESOURCE INDICATORS FOR LAOS

Forest area

71.6

% of land area



▲ better

Tree cover loss

318k

hectares



▼ worse

Protected areas

19.0

% of territory



▲ better

Resource depletion

3.44

% of GNI



▼ worse

Net forest depl.

1.59

% of GNI



▼ worse

Freshwater use

3.86

% of internal resources



▲ better

Regional reference is the MEDIAN of the 8 countries, never the mean. Missing values are shown as "no data", never 0. Ranks are direction-adjusted, so rank 1 is always the best outcome.

Source: World Bank Sovereign ESG · esgdata.worldbank.org

Figure 4. Design B: one click updates the KPI strip, small multiples and trend together.

VIEW

Map + ranking



% of land area · World Bank ESG

8 countries · annual observations
1990–2022, coverage varies

Gray hatching = no data (never zero)

REGIONAL DISTRIBUTION

Country ranking

Selected year · higher values shown first

2022



REGIONAL AVERAGE

46.8%

Unweighted mean

REGIONAL MEDIAN

46.5%

Midpoint of 8 peers

VIETNAM VS REGIONAL AVERAGE

Near the regional center in 2022

+0.5 pp

Figure 5. Design C, level 1: controls and a ranking of all eight countries, with average and median cards.



click an indicator card

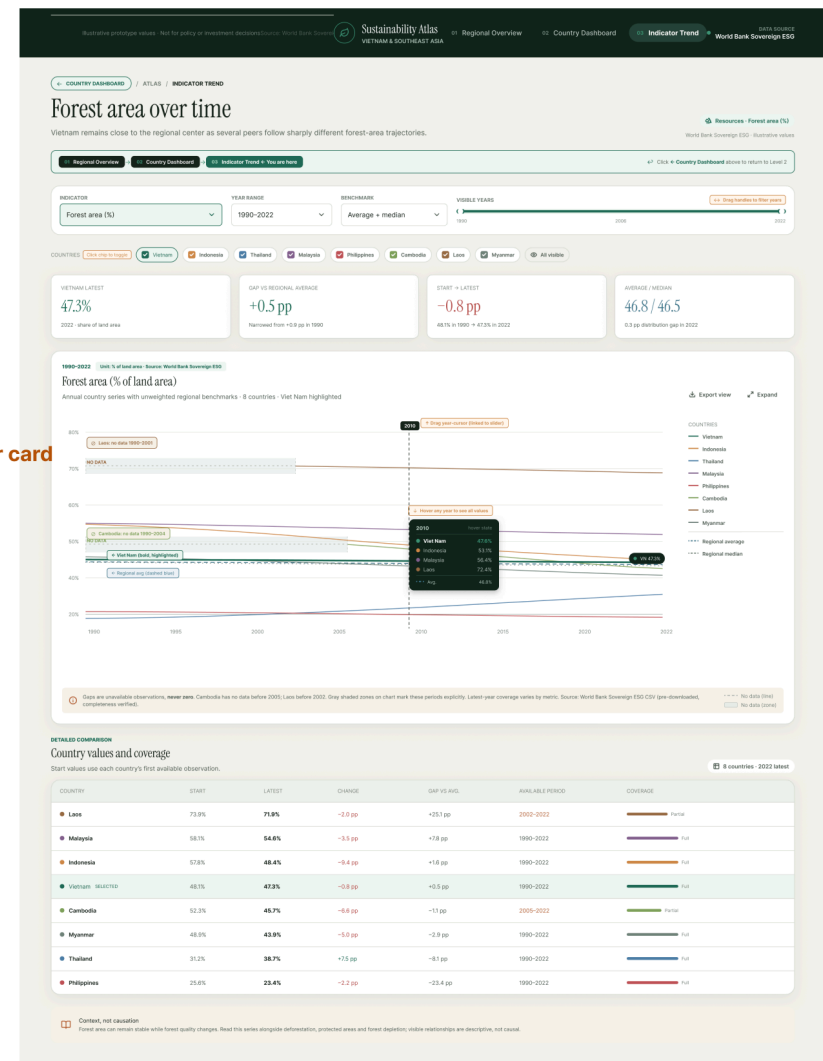


Figure 6. Design C: clicking an indicator card opens its full trend; gaps are named, never drawn as zero.